

Dynamic Waypoint Navigation and Control of Light Weight Powered Paraglider

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Abstract— A Powered Paraglider, also known as Paramotor, has a ram-air inflated canopy in the shape of an aerofoil from which a payload, commonly called as Gondola, housing both propulsion system and control mechanism is suspended. It can lift heavy loads, is quick to setup for rapid launch, and is compact and light-weight, thereby making it ideal for military operations like tactical surveillance and cargo deployment. Paramotors are suitable for scenarios where stable and low speed flying capabilities are necessary. This paper presents a software architecture for guidance and control of light weight small scale Paramotors. For heading and altitude tracking, the system uses feedback compensated control laws. First, linear models are derived that describe both the Paramotor's longitudinal and lateral dynamics. Then, a six degree-of-freedom model is used to describe dynamics, weight, aerodynamic forces on payload and parafoil, aerodynamic moments, effect of apparent forces and moments, moments generated on the centre of mass by the forces exerted at the payload and parafoil. Then system identification based on simplified linear lateral and longitudinal models is used. These simplified linear models are used for designing control laws using classical frequency domain techniques. MATLAB/Simulink was used to simulate the performance of the proposed Paramotor controllers. It was found that the described approach is robust enough for designing control strategies to maintain stability in event of disturbances.

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1. INTRODUCTION

A Powered Paraglider, also referred to as Paramotor, consists of a ram-air inflated canopy in the shape of an aerofoil from which a payload casing the propulsion and control mechanism is suspended (Figure 1). It has high load lifting capacity, fast setup time for rapid launches, highly compact and light-weight, making it useful in military operations such as tactical surveillance and cargo deployment. This type of air vehicle can be successfully used in applications where stable, low speed flying and mobility are required. The powered paraglider was mostly used for recreational purposes so there are very few published works can be found on the current topic[1]–[3]. The current research will focus on developing a guidance control system for a light weight powered paraglider. The aerodynamics parameter of the parafoil will be examined using experimental data as well as data derived using simulation software such as MathWorks's MATLAB - SIMULINK.



Figure 1. Parafoil -Payload System

All conventional and gliding parachute systems have a wide number of applications, ranging from just a few pounds to thousands of pounds, for military, science and civil purposes. Applications of military airdrops include the distribution of personnel, supplies and battle gear and the re-provision of troops to remote areas which are inaccessible. In addition, these are operated to carry ammunition, counter measurements and sensors. Parachute devices technical uses include recuperation of man-made satellites, retrieval of the sounding rockets and satellite booster parts, recuperation of high altitude balloon payloads[4]. Military uses cover humanitarian assistance airdrop, sporting parachuting and the stability and deceleration of the ejection seat. Additional proposed use of powered parafoils can include aerial spraying of different pesticides, slow-moving, low-altitude photography platforms, ground-penetration radar observation of anomalies[5]. It can assist as an ideal platform for missions requiring long endurance times such as radio frequency relay.

Parafoil-load devices are special recuperation, rescue and freight distribution instruments; currently, they are used in many different applications, for several reasons:

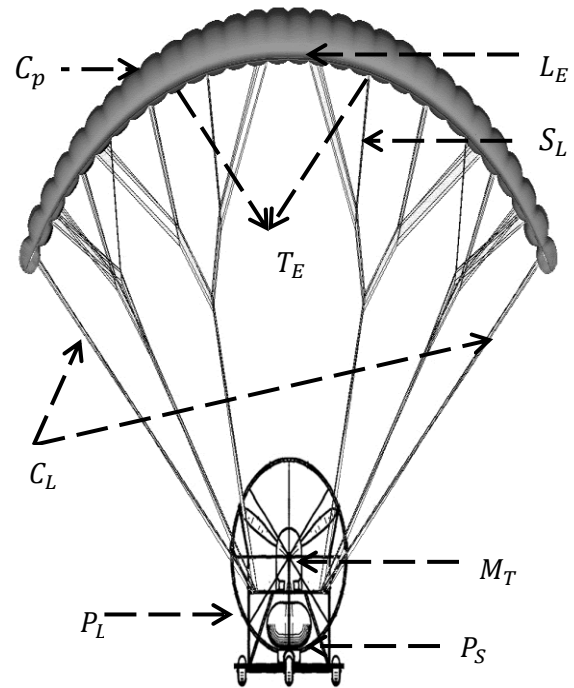
- Affordable.
- Small and compact, which renders them critical to air supply and recovery storage
- Precise and quick landing allows the parafoil load method useful for moving critical equipment and people injured.

A mathematical model is developed using conventional method of analytical dynamics. Assuming the model as a single rigid body a 6-DOF mathematical model that is suitable for detailed simulation is developed. The 6-DOF mathematical model for the light weight powered paraglider is used for simplified, linear longitudinal and lateral models that are then solved using MATLAB.

2. EQUIPMENT AND DESIGN

Parafoil Performance Parameters

Parafoils have two lateral and longitudinal degrees of freedom. Two methods may be used for lateral (directional) control. First, the shortening of the suspensions on the canopy edge leads to a deflection in the direction of the anticipated turn. The lift on the deflected hand reduces and induces a tilt or a roll [6]. The second method uses the asymmetrical deflection of the parafoil's trailing edge. The consequence of a deflection of an asymmetric flap depends on the extent of the deflection and incidence of the canopy angle.



C_p : Canopy
 T_E : Trailing edge
 C_L : Control Lines
 M_T : Motor
 L_E : Leading Edge
 S_L : Suspension Lines
 P_L : Gondola/Payload
 P_S : Power Supply

Figure 2 Paraglider / Ram air Parachute Main Parts[7]

Longitudinal control can also be achieved by two methods. First, by adjusting the rigging angle ϕ as shown in Figure 3, either increasing or decreasing suspension lines lengthwise. This approach is best suited to small parafoils, improper rigging may result in a static solid, leading edge collapsing, and a loss of lateral stability. The second longitudinal control method uses pitch control, throttle with speed control and symmetrical flap deflections.

Paraglider steering Mechanism

Two specific methods are available to steer a paraglider. Direct control of the servo and regulation of steering bars. The design is not practical to use other steering methods such as thrust vectoring or multiple propellers.

Direct Control Mechanism - Direct control mechanisms are based on servo motors that are connected directly to the parafoil control rods. The servo motors are placed at several corners which drive the control lines connected to the to the parafoil. Direct servo control is only used if the servos can redirect the parafoil side to the appropriate distance to allow the parafoil to turn a small circle. We in our flight have used direct servo control, hence we would not concentrate on steering bar control mechanism.

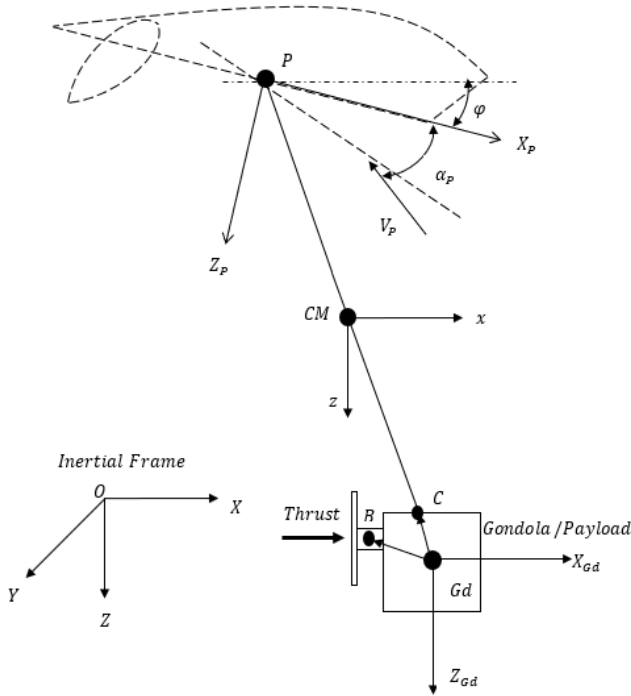


Figure 3 Parafoil and payload schematics [7]

3. PARAFOIL-PAYLOAD SYSTEM MATHEMATICAL MODEL

The design for a paramotor described in this paper is for an extensive range of parafoils, payloads and aerodynamic parameters with a different size, mounting, and turning speeds. The momentary response due to a change of angle of attack is small and it damps easily compared to the time needed to change deflection of the flap due to flap deflecting control servos. Airborne ram aircraft can be mathematically represented in two groups approximatively. First class consists of observing the relative motion of the parafoil and payload. The second classification calls the whole process a static structure, which lacks subjective motion in the layout of the control system.

Generally, a 6 degree-of-freedom (DOF) represents location and angle of the paramotor, 2 and 3 DOF is to design relative position of payload to parafoil. [8], [9] exploited a technique for video measurements between parafoil and unpowered or glided parafoil system to measure relative orientation. Hur and Valasek [10] added multiple sensors on the parafoil to evaluate the 8 DOF design for the efficient motorized buckeye system.

From [1] presented that the predictive control method is the efficient way to check the unpowered or gliding parafoil's horizontal path while Toglia [11] built and tested an algorithm for parafoil payload systems through simulations. Chambers research (2007) [12] focused on the control of longitudinal and altitude for small paramotors. In this segment we define a complex model that displays the longitudinal and the lateral movement of a control and guidance device. The system under consideration compromise a parafoil-payload system controlled by the control lines.

Kinematics and Dynamics

A 6DOF dynamic model has been developed to simulate the paramotor motion for controller design and validation. The paramotor is modelled as one solid structure, although there can be a significant relative motion between the canopy and fuselage. Figure 3 displays the geometry of the parafoil-payload system. The 6 DOF consists of 12 component state vectors, including the location of the global center of mass (CM), $\{[x, y, z], [\phi, \theta, \Psi], [u, v, w], [p, q, r]\}$.

Balance of Forces

About the center of mass, the equation is written as

$$m \frac{dV_{CM}}{dt} + m \Omega V_{CM} = W + F_A^P + F_A^{Gd} + F_T + F_{App} \quad (1)$$

Where,

m	Total mass	F_A^{Gd}	Aerodynamic force (payload)
W	Weight Force	F_T	Thrust
F_A^P	Aerodynamic force (parafoil)	F_{App}	Apparent force

The force occurs in the middle of the mass and must be transferred onto the body's defined reference frame. g is acceleration due to gravity. The thrust force F_T operates in the positive X direction of the gondola / payload system via motor location B . The only aerodynamic working on the payload is the drag at the centers of payload and is generated by Eqn. (3), because the gondola / payload cannot produce lift. Lift and drag are both component of the aerodynamic forces on the parafoil. Eqn. (4) provides the net force that works on the parafoil. F_{App} is the apparent force vector of the body Eqn. (5) due to the fluid motion around the object.

$$W = m [T][0, 0, g] \quad (2)$$

$$F_A^{Gd} = -\frac{1}{2} \rho S_{Gd} |V_{Gd}| C_D^{Gd} \begin{bmatrix} u_{Gd} \\ v_{Gd} \\ w_{Gd} \end{bmatrix} \quad (3)$$

$$F_A^P|_{parafoil} = \frac{1}{2} \rho S_P |V_P| \left(C_L^P \begin{bmatrix} w_P \\ 0 \\ -u_P \end{bmatrix} - C_D^P \begin{bmatrix} u_P \\ v_P \\ w_P \end{bmatrix} \right) \quad (4)$$

$$F_{App} = -m_{App} a_{CM} - \Omega m_{App} V_P \quad (5)$$

The resultant aerodynamic force in the body frame of parafoil is given by

$$F_A^P = [T]^{BP} F_A^P|_{parafoil} \quad (6)$$

The transformation matrix given by $[T]$ and $[T]^{BP}$ are given as

$$[T] = \begin{bmatrix} C(\Psi)C(\theta) & S(\Psi)S(\theta) & -S(\theta) \\ C(\Psi)S(\theta)S(\Phi) - S(\Psi)C(\Phi) & S(\Psi)S(\theta)S(\Phi) + C(\Psi)C(\Phi) & C(\theta)S(\Phi) \\ C(\Psi)S(\theta)S(\Phi) + S(\Psi)S(\Phi) & S(\Psi)S(\theta)C(\Phi) - C(\Psi)S(\Phi) & C(\theta)C(\Phi) \end{bmatrix}$$

$$[T]^{BP} = \begin{bmatrix} \cos \varphi & 0 & \sin \varphi \\ 0 & 1 & 0 \\ -\sin \varphi & 0 & \cos \varphi \end{bmatrix}$$

Moment equilibrium

Moment equilibrium of the center of mass [13]–[16]

$$\begin{aligned} I\dot{\omega} + I\Omega\omega &= M_A^P + M_{App} + R_{CM-P}F_A^P \\ &\quad - R_{CM-P}m_{App}\dot{V}_{CM} \\ &\quad - R_{CM-P}\Omega m_{App}V_P \\ &\quad + R_{CM-Gd}F_A^{Gd} + R_{CM-B}F_T \end{aligned} \quad (7)$$

Where,

M_A^P : Aerodynamic moment, M_{App} : Apparent Moment

$R_{CM-P}F_A^P, R_{CM-Gd}F_A^{Gd}, (R_{CM-P}m_{App}\dot{V}_{CM} + R_{CM-P}\Omega m_{App}V_P), R_{CM-B}F_T$ are moments due to the motor displacement from the mass center of the system [13]–[16]

$$M_A^P = \frac{1}{2}\rho S_P |V_P|^2 \begin{bmatrix} \frac{C_{l_p}}{2|V_P|} b^2 p + C_{l_\phi} b \phi \\ \frac{C_{m_q} c^2 q}{2|V_P|} + C_{m_c} \\ \frac{C_{n_r} b^2 r}{2|V_P|} \end{bmatrix} \quad (8)$$

$$M_{App} = -I_{App}\dot{\omega} - \Omega I_{App}\omega - \Lambda_P m_{App} V_P \quad (9)$$

where R_{CM-P} and Λ_P are the skew-symmetric form of X_{CM-P} and V_P

$$R_{CM-P} = \begin{bmatrix} 0 & -X_{CM-P}(1) & X_{CM-P}(2) \\ X_{CM-P}(3) & 0 & -X_{CM-P}(1) \\ -X_{CM-P}(2) & X_{CM-P}(1) & 0 \end{bmatrix}$$

$$\Lambda_P = \begin{bmatrix} 0 & -w_P & v_P \\ w_P & 0 & -u_P \\ -v_P & u_P & 0 \end{bmatrix}$$

Apparent Mass

The canopy of parafoil is lightweight and far back from the mass center of the parafoil system. The motion of parafoils is significantly altered by the apparent mass of the adjacent air. The resulting system movement is as though it had greater weight when the apparent additional mass is due to the forces necessary to speed up the liquid. In their work Lissaman and brown [17] presented an overview of how the apparent mass requirements for flat rigged canopy should be approximated.

Flaps Deflection

The parafoil's trailing edge is regulated by the control lines, deflection is referred to as brake. When they're deflected at the same direction, lift and drag improve but the performance declines. Asymmetrical flap deflection results in a change in the moment that turns the paramotor. Assume δ_L and δ_R to be the left and right brake deflections.

For both situations, δ_s = minimum (δ_L, δ_R) and $\delta_a = (\delta_L - \delta_R)$ represents symmetrical and asymmetrical flap deflection. The force due to brake is given by

$$F_\delta^P = S_{F_\delta^P} \begin{bmatrix} \delta_a \\ \delta_s \end{bmatrix} \quad (10)$$

Directional control is achieved by δ_a , introducing rolling and yawing moments and is given by

$$M_\delta^P = S_{M_\delta^P} \begin{bmatrix} \delta_a \\ \delta_s \end{bmatrix} \quad (11)$$

Where,

$$S_{F_\delta^P} = \frac{q_P}{|V_P|} \begin{bmatrix} (C_{L_{\delta_a}} w_P - C_{D_{\delta_a}} u_P) \text{sign}(\delta_a) & C_{L_{\delta_s}} w_P - C_{D_{\delta_s}} u_P \\ -C_{D_{\delta_a}} v_P \text{sign}(\delta_a) & -C_{D_{\delta_s}} v_P \\ (-C_{L_{\delta_a}} u_P - C_{D_{\delta_a}} w_P) \text{sign}(\delta_a) & -C_{L_{\delta_s}} u_P - C_{D_{\delta_s}} w_P \end{bmatrix}$$

$$M_\delta^P = q_P \begin{bmatrix} C_{l_{\delta_a}} b/t & 0 \\ 0 & 0 \\ C_{n_{\delta_a}} b/t & 0 \end{bmatrix}$$

The equations derived for the paramotor system can be written in matrix for as follows

$$A \begin{bmatrix} \dot{V}_{CM} \\ \dot{\omega} \end{bmatrix} = D + S \begin{bmatrix} \delta_a \\ \delta_s \end{bmatrix} \quad (12)$$

where,

$$A = \begin{bmatrix} m + m_{App} & 0 \\ R_{CM-P}m_{App} & I + I_{App} \end{bmatrix}$$

D can calculated using Eqns. (4) and (10)

$$S = \begin{bmatrix} S_{F_\delta^P} \\ S_{M_\delta^P} + R_{CM-P}S_{F_\delta^P} \end{bmatrix}$$

Matrix A is non-singular and invertible, thus

$$\begin{bmatrix} \dot{V}_{CM} \\ \dot{\omega} \end{bmatrix} = f(V_{CM}, \omega) + g(V_{CM}, \omega) \begin{bmatrix} \delta_a \\ \delta_s \end{bmatrix}$$

Simplified Linear 6 DOF Model

For high fidelity applications, the above 6 DOF method is useful. A linearized equation of motion is developed in conjunction with the dynamics of the parafoil-payload system in order to layout the control system and stability analysis. In [18] author has recognized two steering methods roll and skid steering. By pushing the brakes downward, the lift of the respective side of the parachute decreases and produces a rolling moment. Downwards brake deflection in skid steering mode raises the drag that is greater than the lift which causes a yawing moment. Some basic assumptions taken are

- Moment is only due aerodynamic forces at X_{CM} .
- In equilibrium conditions, apparent mass and inertia effects are marginal
- In comparison to parafoil drag the payload drag is smaller and can be neglected.
- The parafoil center and center on mass move with same velocity.

Above approximation reduces equation of motion to

$$I\dot{\omega} + \Omega I\omega = M_A^P + S_{M_\delta^P} \begin{bmatrix} \delta_a \\ \delta_s \end{bmatrix} \quad (13)$$

Using Taylor series expansion for linearization.

$$F(x) = F(x_0) + J(x) \quad (14)$$

Where $F(x)$: non-linear function, $J(x)$; Jacobian of $F(x)$ at the equilibrium. For steady state flight, θ is constant hence q is zero.

$$\begin{Bmatrix} \dot{\phi} \\ \dot{\psi} \\ \dot{p} \\ \dot{r} \end{Bmatrix} = \begin{bmatrix} 0 & 0 & A_{13} & 0 \\ 0 & 0 & 0 & A_{24} \\ A_{31} & 0 & A_{33} & A_{34} \\ A_{41} & 0 & A_{43} & A_{43} \end{bmatrix} \begin{Bmatrix} \phi \\ \psi \\ p \\ r \end{Bmatrix} + \begin{bmatrix} 0 \\ 0 \\ B_3 \\ B_4 \end{bmatrix} \delta_a \quad (15)$$

Where,

$$\begin{aligned} A_{13} &= 1 & A_{24} &= 1/\cos \theta_0 \\ A_{31} &= \frac{k_1 C_{l_p} b^2 I_{zz}}{2V_p} & A_{41} &= \frac{k_2 C_{l_p} b^2 I_{xz}}{2V_p} \\ A_{33} &= k_1 C_{l_\phi} b I_{zz} & A_{43} &= k_2 C_{l_\phi} b I_{xz} \\ A_{34} &= \frac{-k_1 C_{n_r} b^2 I_{xz}}{2V_p} & A_{44} &= \frac{-k_2 C_{n_r} b^2 I_{xx}}{2V_p} \\ B_3 &= \frac{k_1 b (C_{l_{\delta_a} I_{zz}} - C_{n_{\delta_a} I_{xz}})}{d} & B_4 &= \frac{k_2 b (C_{l_{\delta_a} I_{xz}} - C_{n_{\delta_a} I_{xx}})}{d} \\ k_1 &= \frac{\rho S_P |V_P|^2}{2(I_{xx} I_{zz} - I_{xz}^2)} & k_2 &= \frac{\rho S_P |V_P|^2}{2(I_{xz}^2 - I_{xx} I_{zz})} \end{aligned}$$

The physical characteristics is given in Table 1[7] and aerodynamic parameters are listed in Table 2 [7].

Table 1. Parafoil and Payload physical parameters

Symbol	Value
Mass	1.880 Kg
$\rho(\text{density})$	1.225 kg.m^{-3}
g	9.81 m.s^{-2}
S_P	1.16 m^2
S_{Gd}	0.5 m^2
b	2.15 m
c	0.54 m
d	0.40 m
I (Inertia) kg.m^2	$\begin{bmatrix} 0.0160 & 0 & -0.0065 \\ 0 & 0.0186 & 0 \\ -0.0065 & 0 & 0.0179 \end{bmatrix}$

Table 2. Parafoil and payload system aerodynamic coefficients

Symbol	Value	Symbol	Value
$C_{D_0}^{Gd}$	0.15	C_{l_p}	-0.108
$C_{D_\alpha}^{Gd}$	1.00	C_{l_ϕ}	-0.051
C_{L_0}	0.034	$C_{l_{\delta_a}}$	0.0031
C_{L_α}	2.52	C_{n_r}	-0.087
C_{D_0}	0.077	$C_{n_{\delta_a}}$	0.0043
C_{D_α}	0.277	$C_{L_{\delta_a}}$	0.0001
C_{m_0}	0.016	$C_{D_{\delta_a}}$	0.0001
C_{m_α}	-0.065	$C_{L_{\delta_s}}$	0.21
C_{m_q}	-0.269	$C_{D_{\delta_s}}$	0.30

4. CONTROL ARCHITECTURE

This section describes an autopilot design suitable for the sensors and computational resources available on-board

paramotor. A successive loop closure to design lateral and longitudinal autopilots has been implemented taking into the account the saturation limits. Design process takes into factor that the inner loop has the highest bandwidth, with each successive loop bandwidth a factor of 5 to 10 times smaller in frequency. Each loop has discrete-time implementation of proportion-integral-derivative (PID) feedback control law.

Lateral-directional Autopilot

The linearized lateral dynamics eqn. (15) has been utilized in devising the heading angle tracing controller based on the parameters described in Table 1 and Table 2.

Roll Angle Hold Autopilot - A gyroscope is used to measure the difference in the system's attitude to the reference attitude of $Y - Z$ plane of system. Feedback can be used to adjust the system's rolling angle from the deviation signal for flap deflection. This autopilot is used to provide a set compass heading as inner loop for the autopilot. The layout for roll control is illustrated in Figure 4 [19].

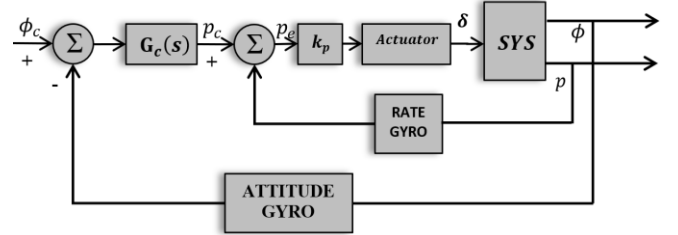


Figure 4. Autopilot for Roll – Angle Hold [19]

Heading Hold Autopilot - This is to keep the plane on a specified heading of the compass. An additional feedback circuit is inserted around the roll-angle feedback circuit. Figure 5 displays the block diagram for the autopilot heading.

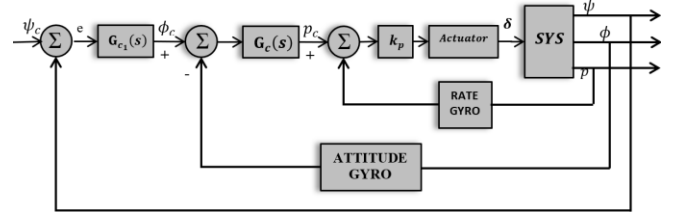


Figure 5. Autopilot for Heading –Hold [19]

The servo mechanism used for regulating is built in the actuator block as the first order function [16].

Longitudinal Autopilot

Airspeed plays a key role in the longitudinal structure of the longitudinal autopilot. Our goal in the development of the longitudinal autopilot was to control the airspeed and altitude via the throttle and brake deflection as an actuator.

Pitch Hold Control Logic- The pitch hold loop is similar to the roll hold loop, and in its evolution, we will follow a similar path.

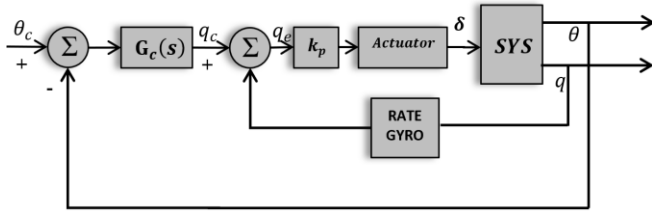


Figure 6. Pitch attitude hold feedback loops [20], [21]

Airspeed Hold Using Throttle Control Logic- The dynamic model for airspeed using throttle is shown in Figure 7. The gains are such that chosen so that the bandwidth of the outer loop is less than the bandwidth of the inner loop. The transfer functions T_{F1} and T_F are derived using equation of motion given in [20], [21].

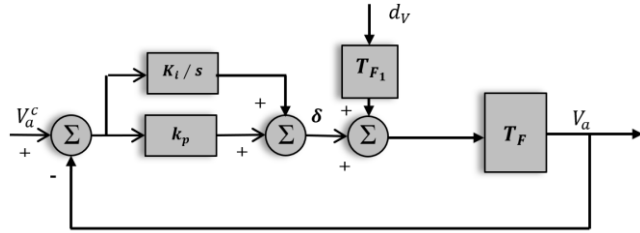


Figure 7. Airspeed hold using throttle [20], [21]

Path Following - The strategy in this section will be to construct a desired course angle at every spatial point relative to the straight-line path that results in the UAV moving toward the path. For path following, the cross-track error e_{py} should subside to zero by commanding the course angle. Straight-line path following problem is to select χ_c so that $e_{py} \rightarrow 0$ when χ_q is known shown by Figure 8.

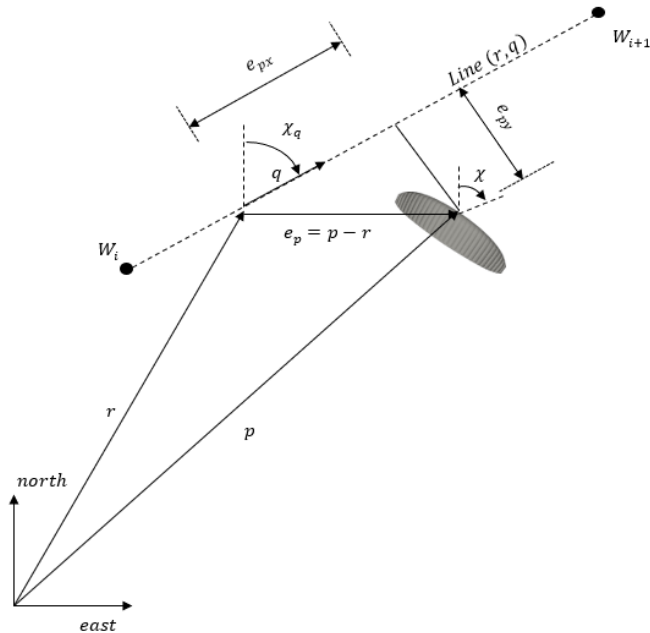


Figure 8. UAV position and orientation represented by (p, χ) , and the straight-line path indicated by $Line(r, q)$.

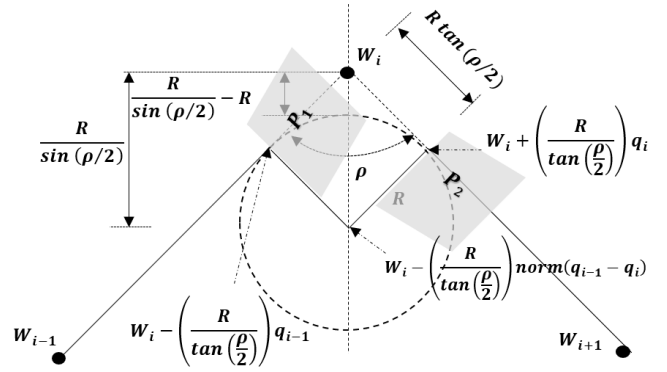


Figure 9. The transition from straight-line path $W_{i-1}W_i$ to W_iW_{i+1} using pre-turn

5. SIMULATION RESULTS

The execution of the complete algorithm has been simulated in MATLAB-SIMULINK environment. The parameters used for the simulation are listed in Table 1 and Table 2. The control architecture used has been explained in section 4. The parafoil-payload system was flown at velocity of 6 m/s, assuming constant density and zero/negligible wind, the algorithm has pre-turn manoeuvre. To show the quality of the control and direction logic, a route of four points has been chosen. Each waypoint had an effective radius and plane P_1 and P_2 Figure 9; if the paramotor reached P_1 then the paramotor is assumed to have reached waypoint and is directed to move in the circular path, similarly on reaching P_2 it is commanded to move to another waypoint. Figure 10 shows the trajectory followed by the paramotor starting at location $(0,0)$ in a North direction. Figure 11 depicts the variation in altitude during the simulation.

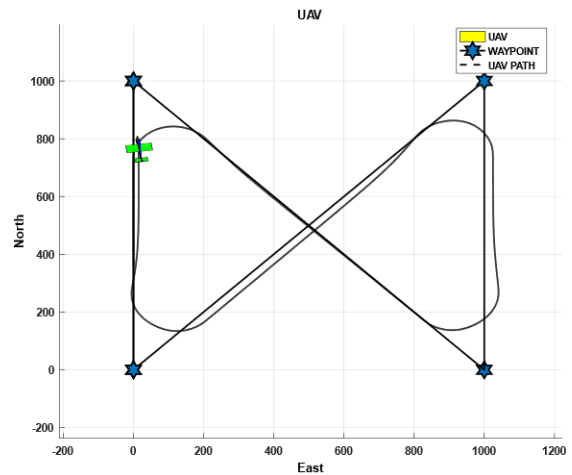


Figure 10. Flight paths of UAV

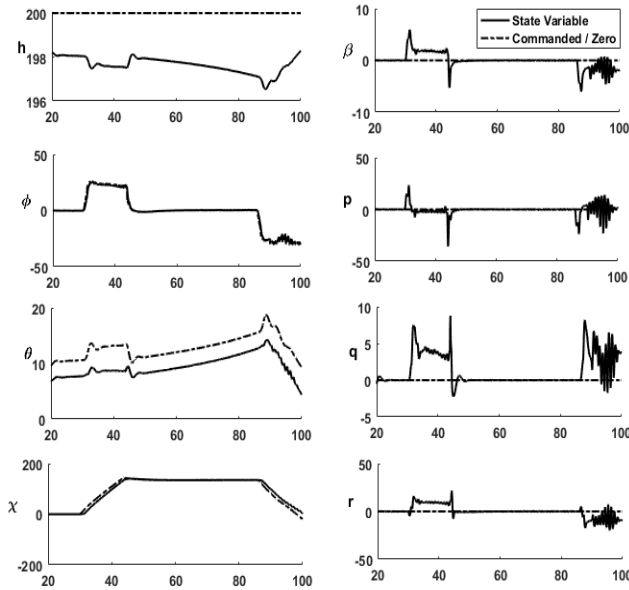


Figure 11. State Variable of Waypoint Navigation.

6. CONCLUSION

This paper presented a detailed simulation, parameter identification and small-scale paramotor controller development model. A detailed 6DOF non-linear mathematical model was derived using some assumptions. The non-linear 6-DOF model was then converted to low fidelity model to create a set of linear equations explaining the paramotor's lateral dynamic and appropriate for control development. Similarly, simplistic longitudinal dynamic models were also developed. Successive loop closing techniques have been developed to improve stability using the highly practical feedback technique. Lastly, it was demonstrated by simulation that lateral and longitudinal paramotor control could be successfully decoupled by using asymmetric brake deflection for lateral control and maintaining altitude and speed control by means of throttle and pitch control. Further work involves using SIL (software-in-loop) and HIL (Hardware-in-loop) simulations to validate in control architecture. Then implement the control law to real flight.

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